Set 2E: Slew-Rate Limits

ECEN 325 — Lecture 2 (Operational Amplifiers) practice set

What this set trains. An op-amp output cannot change faster than its slew rate SR. For a sinusoid of amplitude $\hat{V}$ at ω the fastest change is $\omega\hat{V}$, so every op-amp output must satisfy $\omega\hat{V} \le SR$. You find the gain from V_i to every op-amp output at the input frequency, pick the largest one, and solve for the largest input amplitude. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-2 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1 tools for impedances, dividers, Bode plots and sinusoids):

T1 *Ideal op-amp with negative feedback:* no current into + or −; $V_- = V_+$ (virtual short). Its output is an ideal voltage source (loads on it do not matter).

T2 *Block formulas:* inverting $\frac{V_o}{V_{in}} = -\frac{Z_f}{Z_{in}}$; non-inverting $\frac{V_o}{V_+} = 1 + \frac{Z_f}{Z_g}$ (with V_+ from the divider at the + input); a follower has gain 1. Several blocks in cascade: multiply.

T3 *KCL with impedances:* at a node, $\sum_k \frac{V_{node} - V_k}{Z_k} = 0$ (for a capacitor $\frac{1}{Z} = sC$). Write it at every node except op-amp outputs and source nodes.

T4 *Solving KCL equations:* pick the equation with the fewest unknowns, solve it for one node voltage, substitute into the rest; repeat until only V_o and V_i remain.

T5 *Finite gain:* $V_{out} = A(s)(V_+ - V_-)$, $A(s) = \frac{A_0}{1+s/\omega_b}$, $\omega_t = A_0\omega_b$. With $V_+ = \alpha_+ V_{in}$, $V_- = \alpha_- V_{in} + \beta V_{out}$: $\frac{V_{out}}{V_{in}} = \frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$.

T6 *Slew rate:* for $V_{out} = \hat{V}\sin\omega t$, the fastest change is $\left|\frac{dV_{out}}{dt}\right|_{max} = \omega\hat{V}$; no distortion requires $\omega\hat{V} \le SR$.

Steps:

1. *Sources.* DC and AC both present ⇒ superposition.
2. *DC circuit:* $V_i \to 0$, $C \to$ open; pass the DC through each block with its gain at $s = 0$.
3. *Op-amp rules:* confirm negative feedback; apply T1 (or T5 for a non-ideal op-amp).
4. *Transfer function:* standard blocks → T2; anything else → T3 + T4.
5. *Standard form*, 6. *Bode*, 7. *Output* — exactly as in Lecture 1.

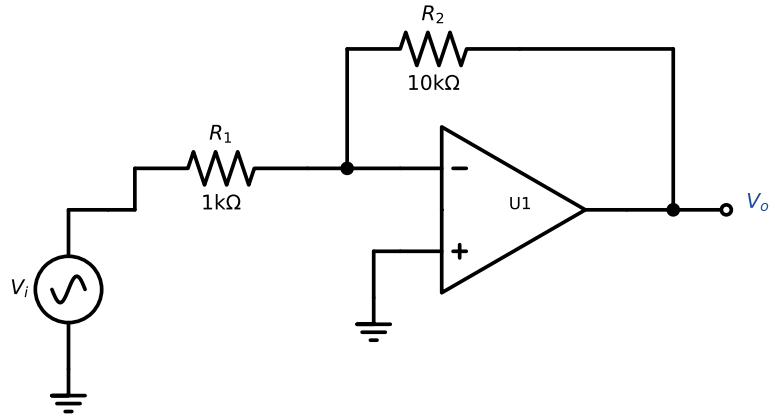
Variations: a design problem runs these steps backwards (Bode plot → $H(s)$ → blocks → component values); a slew-rate problem stops after Step 4 for every op-amp output and applies T6.

Problems

Easy problems

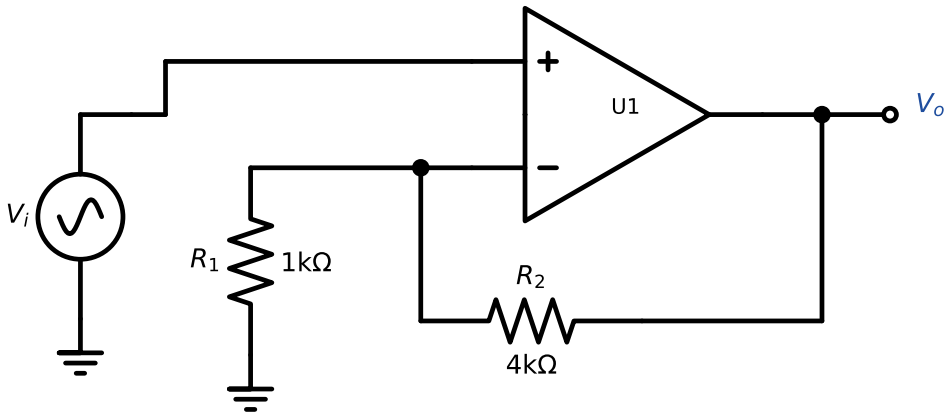
Problem 2E.1 — Easy

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.5\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 20000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



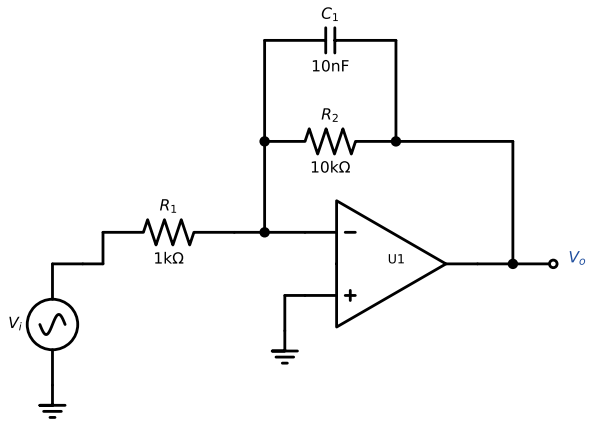
Problem 2E.2 — Easy

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 1\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 50000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



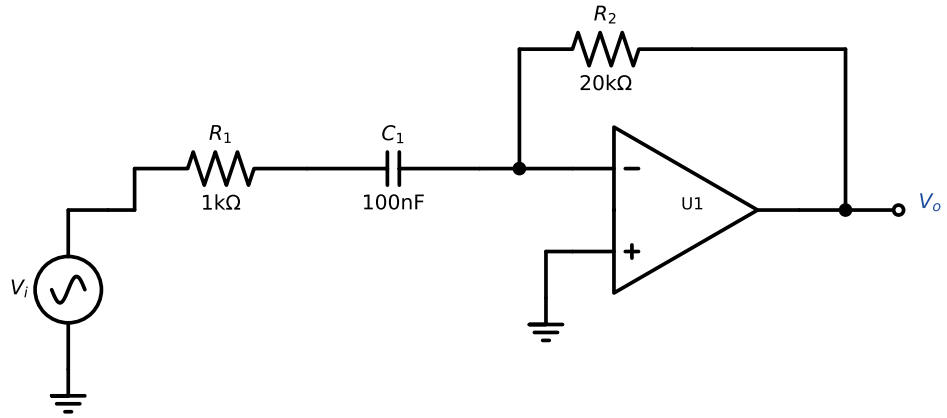
Problem 2E.3 — Easy

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.2\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



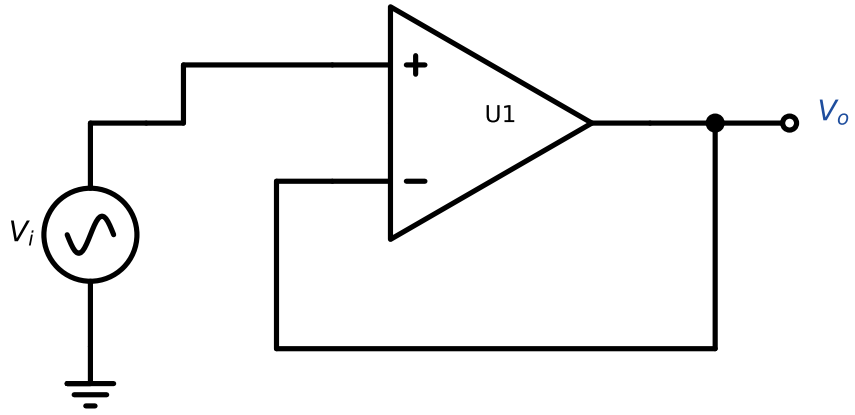
Problem 2E.4 — Easy

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 2\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



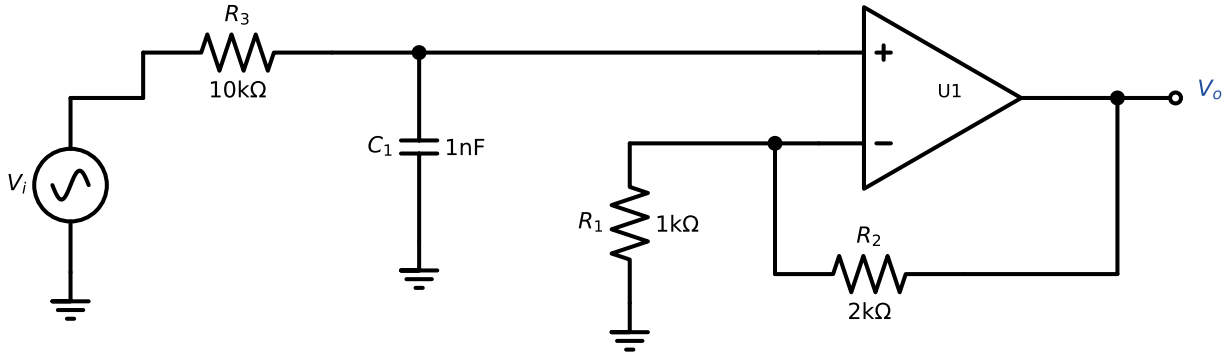
Problem 2E.5 — Easy

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.3\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10^5\, t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



Problem 2E.6 — Easy

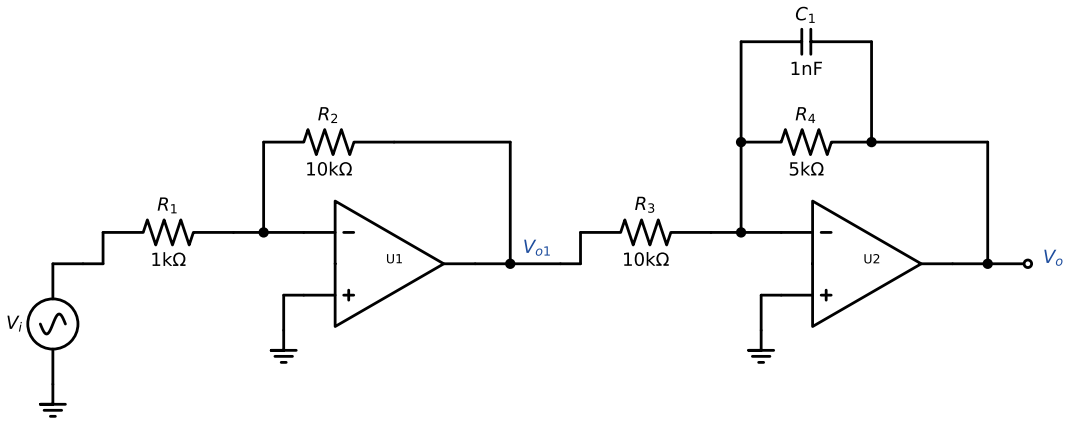
In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 1\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 20000\, t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



Medium problems

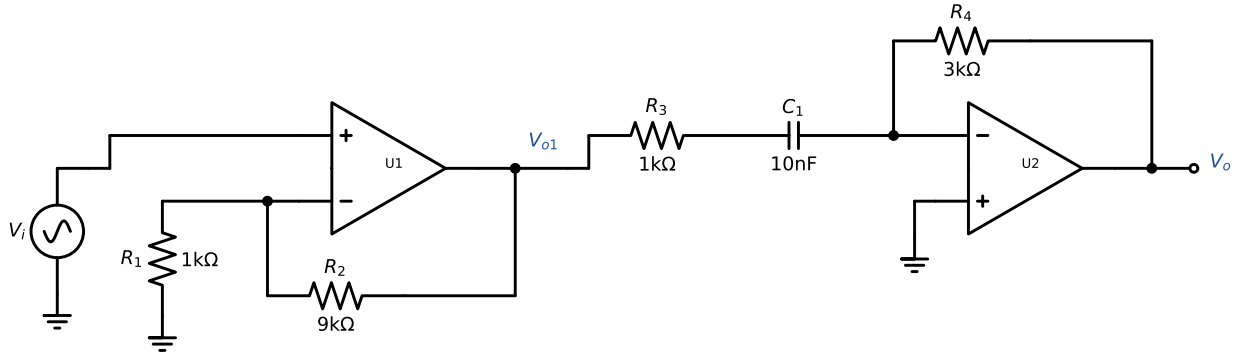
Problem 2E.7 — Medium

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 1\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 50000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



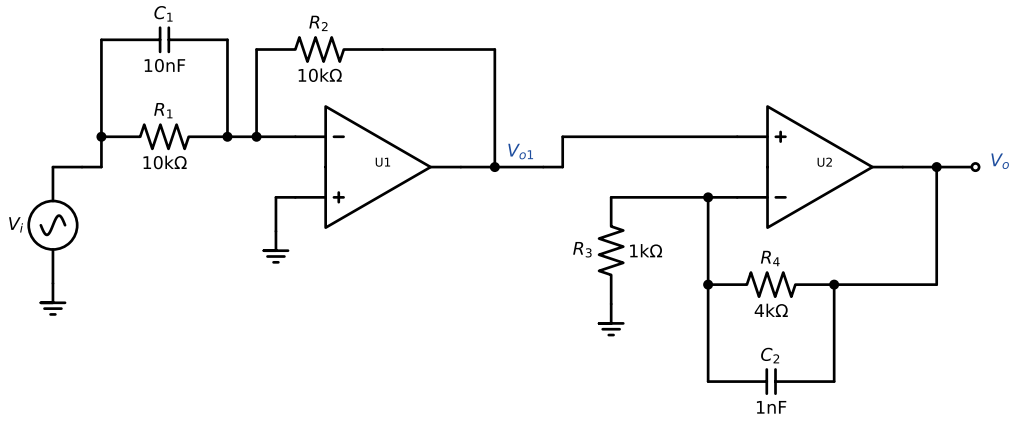
Problem 2E.8 — Medium

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.5\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 3000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



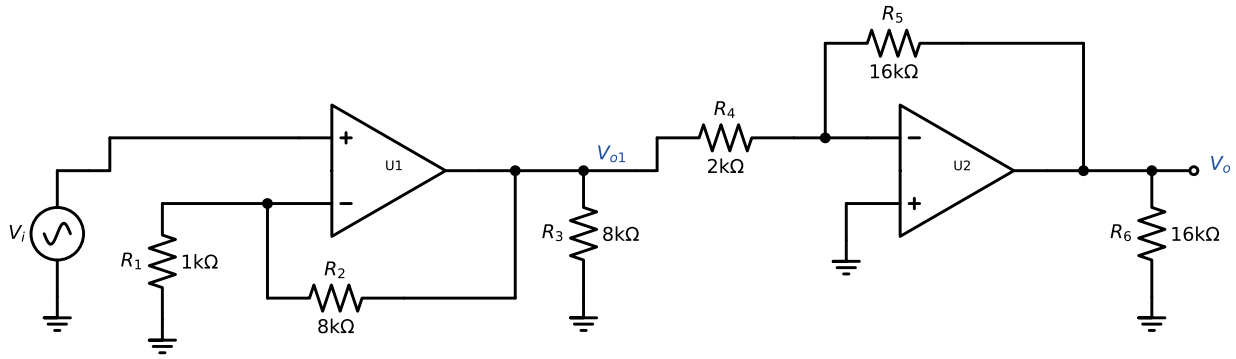
Problem 2E.9 — Medium

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 1\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



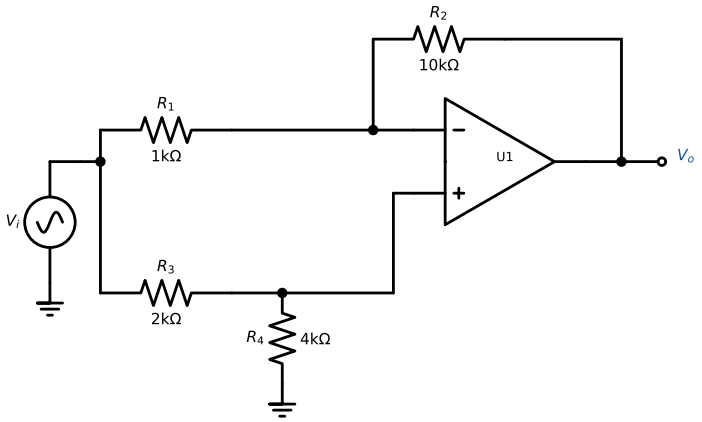
Problem 2E.10 — Medium

In the circuit below (op-amps with open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 20}}$), every op-amp has the slew rate $SR = 2\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 2 \times 10^5\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



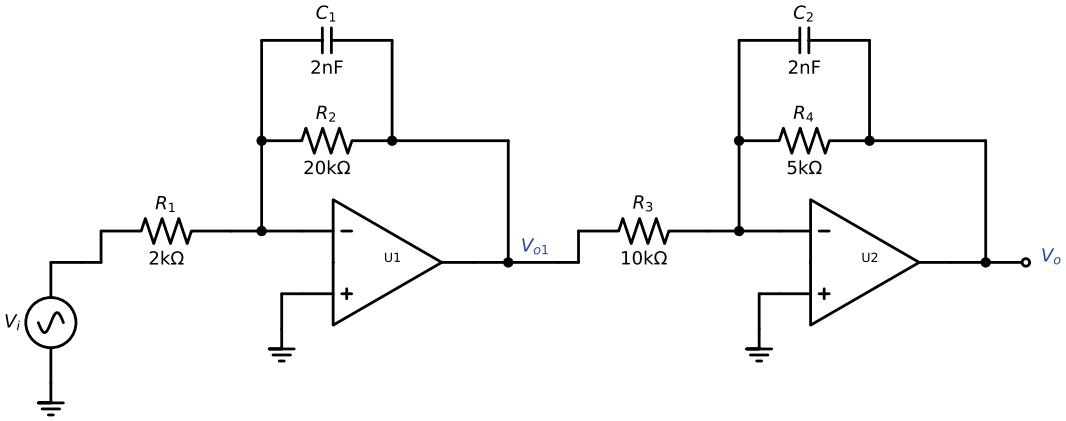
Problem 2E.11 — Medium

In the circuit below (op-amps with open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$), every op-amp has the slew rate $SR = 3\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 50000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



Problem 2E.12 — Medium

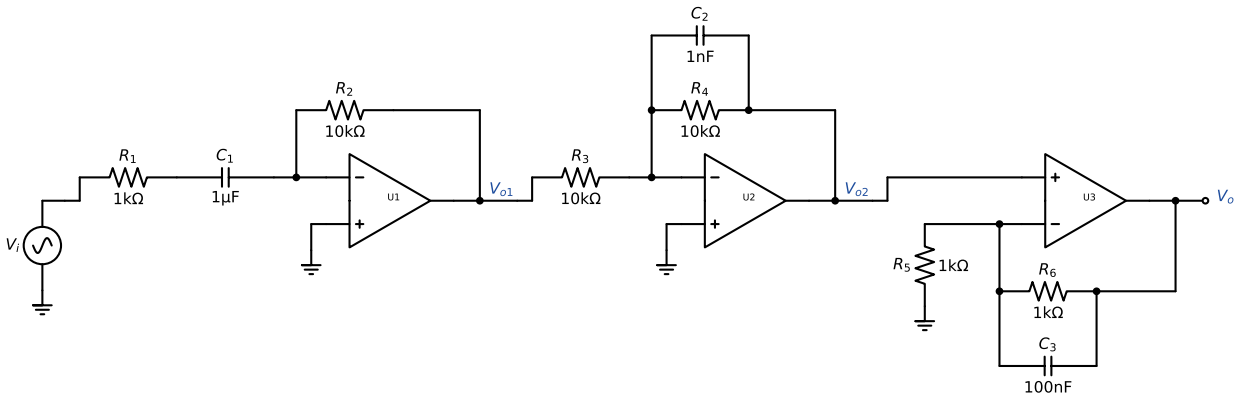
In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.5\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 8000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



Hard problems

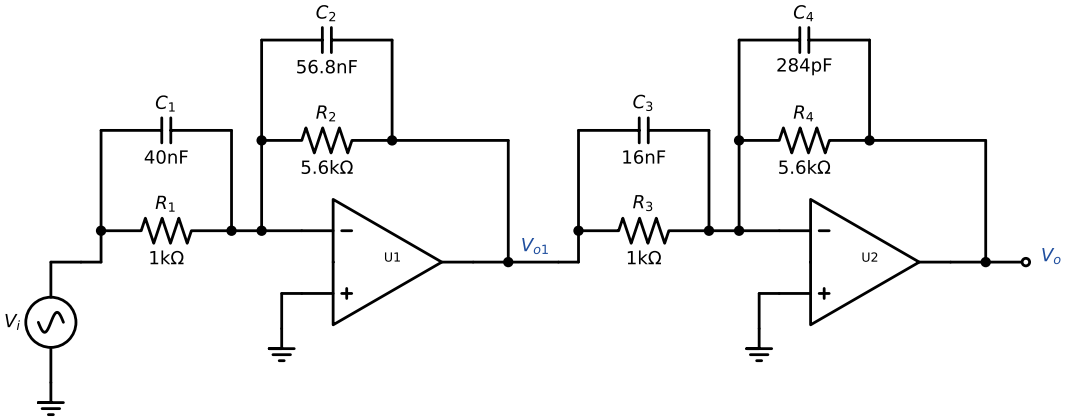
Problem 2E.13 — Hard

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 1\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 20000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



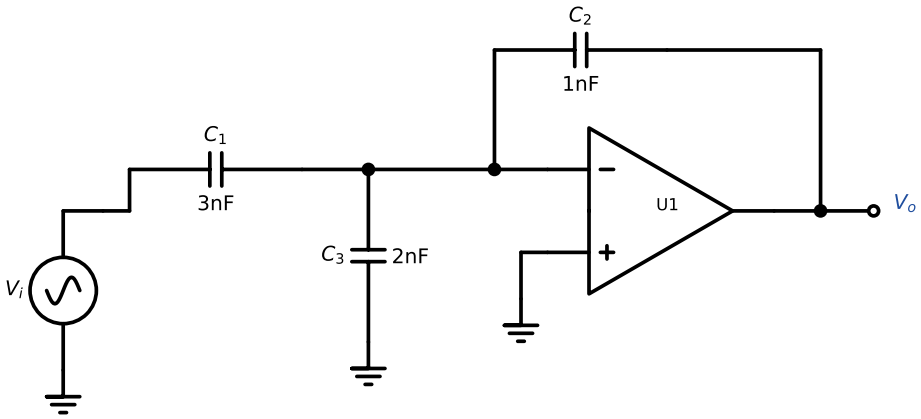
Problem 2E.14 — Hard

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.5\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 2000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



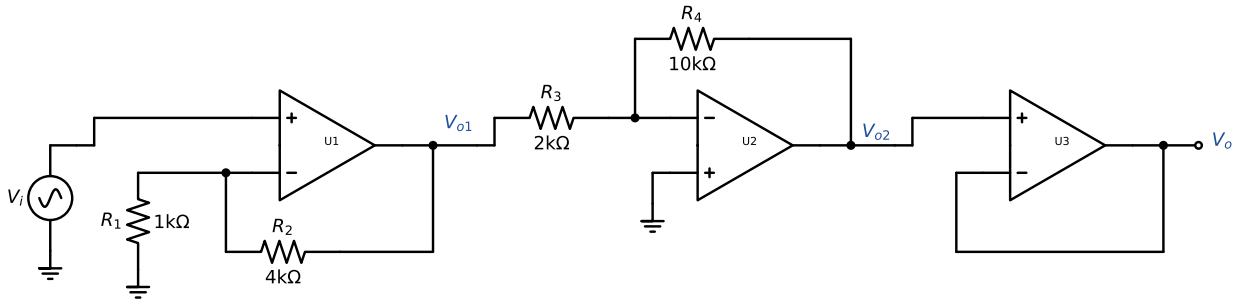
Problem 2E.15 — Hard

In the circuit below (op-amps with open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 30}}$), every op-amp has the slew rate $SR = 5\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 5 \times 10^5\, t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited.



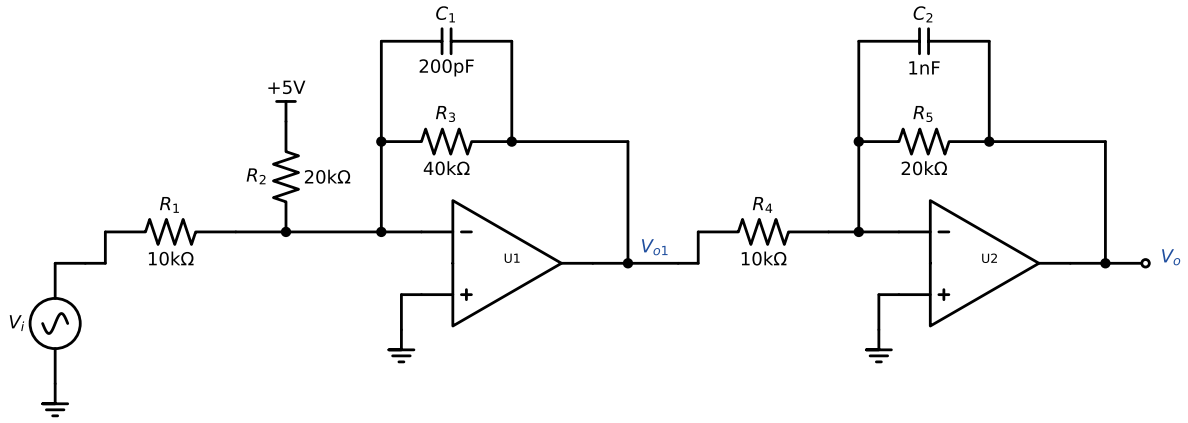
Problem 2E.16 — Hard

In the circuit below (op-amps with open-loop gain $A(s) = \dfrac{10^5}{1 + \dfrac{s}{2\pi \cdot 10}}$), every op-amp has the slew rate $SR = 2\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10^5\, t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



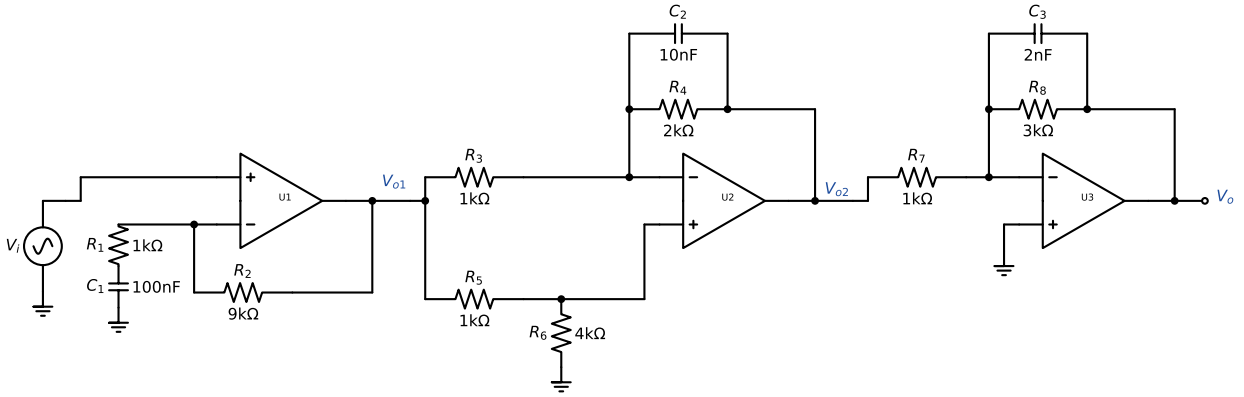
Problem 2E.17 — Hard

In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.2\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 10000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



Problem 2E.18 — Hard

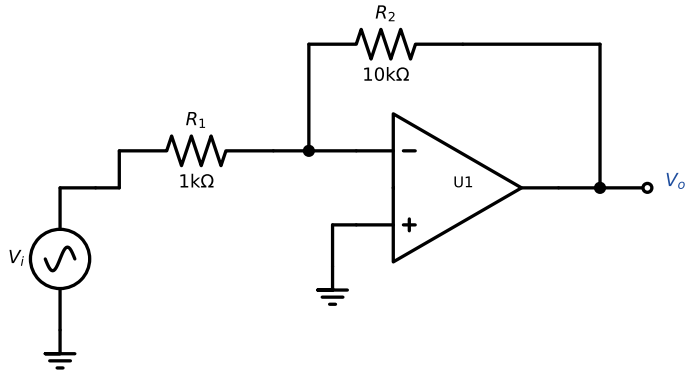
In the circuit below (ideal op-amps (except for the slew rate)), every op-amp has the slew rate $SR = 0.8\,\mathrm{V}/\mu\mathrm{s}$. The input is $V_i(t) = A\sin(2\pi \cdot 6000\,t)$. Find the largest amplitude A for which no op-amp output is slew-rate limited, and say which op-amp output distorts first.



Solutions

Every solution follows the steps of the Lecture-2 pipeline card.

Solution 2E.1 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 20000 = 1.26 \times 10^5$ rad/s and $SR = 0.5\,\text{V}/\mu\text{s} = 5 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k} |\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through R_2: $V_o = 0 - R_2 \cdot \frac{V_i}{R_1}$.

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1} = -10$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 10$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	10	$10A$	$1.257 \times 10^6 A$

There is only one op-amp output.

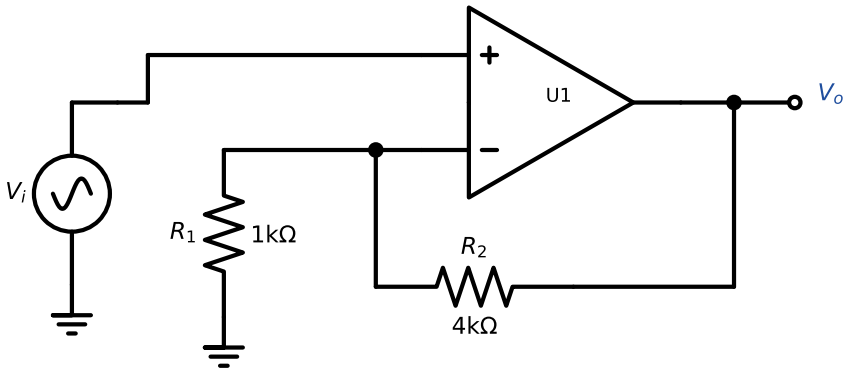
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{5 \times 10^5}{1.26 \times 10^5 \times 10} = 398\,\text{mV}$$

Answer

$A_{max} = 398\,\text{mV}$ (input amplitude); at that point V_o has amplitude 3.98 V.

Solution 2E.2 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 50000 = 3.14 \times 10^5$ rad/s and $SR = 1\,\text{V}/\mu\text{s} = 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The op-amp forces $V_- = V_+$, and R_1, R_2 divide the output: $V_- = \frac{R_1}{R_1+R_2}V_o$, hence $V_o = \left(1 + \frac{R_2}{R_1}\right)V_+$.

$$\frac{V_o}{V_i} = \left(1 + \frac{R_2}{R_1}\right) = 5$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 5$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	5	$5A$	$1.571 \times 10^6 A$

There is only one op-amp output.

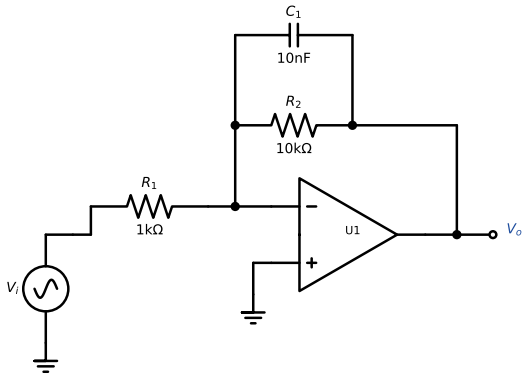
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{10^6}{3.14 \times 10^5 \times 5} = 637\,\text{mV}$$

Answer

$A_{max} = 637\,\text{mV}$ (input amplitude); at that point V_o has amplitude 3.18 V.

Solution 2E.3 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 10000 = 62800\,\text{rad/s}$ and $SR = 0.2\,\text{V}/\mu\text{s} = 2 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k} |\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_o = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 10000}$$

$$\frac{V_o}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-10^5}{s + 10000}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 1.572$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	1.572	$1.572A$	$98760A$

There is only one op-amp output.

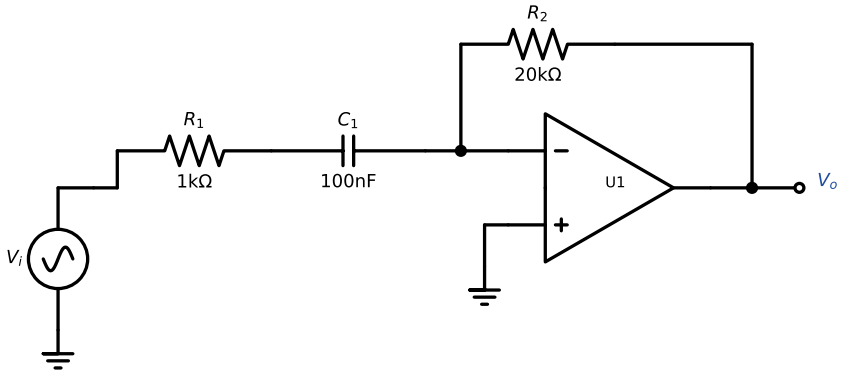
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{2 \times 10^5}{62800 \times 1.572} = 2.03\,\text{V}$$

Answer

$A_{max} = 2.03\,\text{V}$ (input amplitude); at that point V_o has amplitude $3.18\,\text{V}$.

Solution 2E.4 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 10000 = 62800\,\text{rad/s}$ and $SR = 2\,\text{V}/\mu\text{s} = 2 \times 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_o = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

$$\frac{V_o}{V_i} = -\frac{R_2}{Z_{in1}} = \frac{-20\,s}{s + 10000}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 19.75$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	19.75	$19.75A$	$1.241 \times 10^6 A$

There is only one op-amp output.

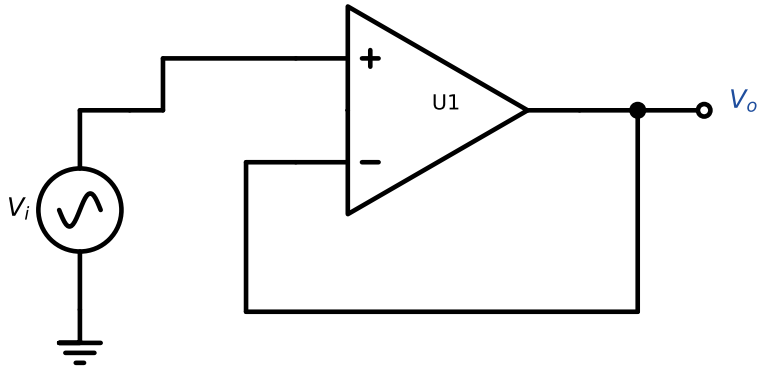
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{2 \times 10^6}{62800 \times 19.75} = 1.61\,\text{V}$$

Answer

$A_{max} = 1.61\,\text{V}$ (input amplitude); at that point V_o has amplitude $31.8\,\text{V}$.

Solution 2E.5 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 10^5 = 6.28 \times 10^5$ rad/s and $SR = 0.3\,\text{V}/\mu\text{s} = 3 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The output is tied back to the − input, so $V_o = V_- = V_+$ (voltage follower).

$$\frac{V_o}{V_i} = 1 = 1$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 1$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	1	$1A$	$6.283 \times 10^5 A$

There is only one op-amp output.

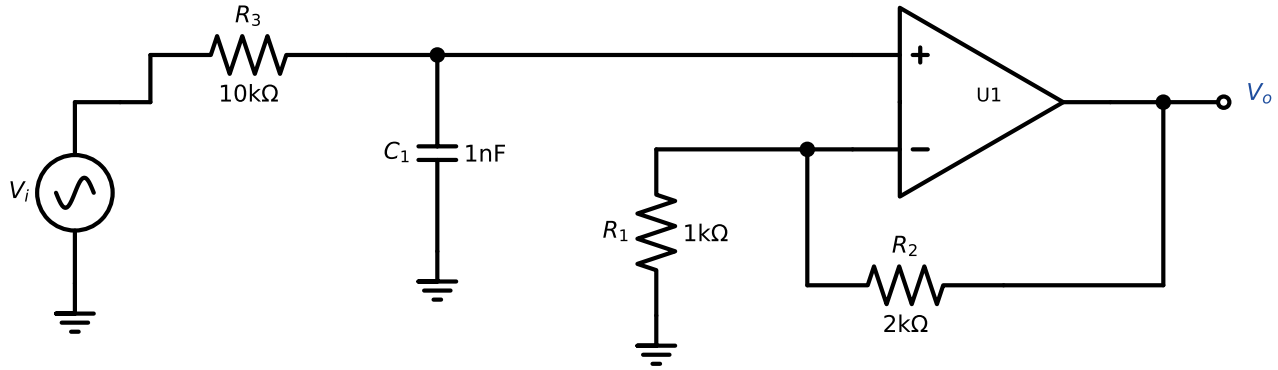
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{3 \times 10^5}{6.28 \times 10^5 \times 1} = 477\,\text{mV}$$

Answer

$A_{max} = 477\,\text{mV}$ (input amplitude); at that point V_o has amplitude 477 mV.

Solution 2E.6 **Easy**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 20000 = 1.26 \times 10^5$ rad/s and $SR = 1\,\text{V}/\mu\text{s} = 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Non-inverting block: No current flows into the + input, so $\frac{1}{sC_1}$ and R_3 form a plain voltage divider: $V_+ = \frac{\frac{1}{sC_1}}{R_3+\frac{1}{sC_1}}V_i$. The op-amp forces $V_- = V_+$, and R_1, R_2 divide the output: $V_- = \frac{R_1}{R_1+R_2}V_o$, hence $V_o = \left(1 + \frac{R_2}{R_1}\right)V_+$.

$$\frac{V_o}{V_i} = \frac{\frac{1}{sC_1}}{R_3 + \frac{1}{sC_1}}\left(1 + \frac{R_2}{R_1}\right) = \frac{3 \times 10^5}{s + 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 1.868$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	1.868	$1.868A$	$2.347 \times 10^5 A$

There is only one op-amp output.

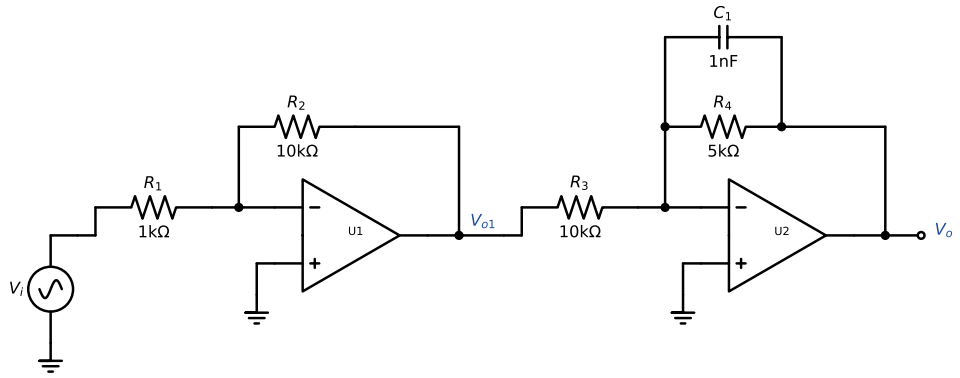
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{10^6}{1.26 \times 10^5 \times 1.868} = 4.26\,\text{V}$$

Answer

$A_{max} = 4.26\,\text{V}$ (input amplitude); at that point V_o has amplitude $7.96\,\text{V}$.

Solution 2E.7 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 50000 = 3.14 \times 10^5$ rad/s and $SR = 1\,\text{V}/\mu\text{s} = 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through R_2: $V_{o1} = 0 - R_2 \cdot \frac{V_i}{R_1}$.

$$\frac{V_{o1}}{V_i} = -\frac{R_2}{R_1} = -10$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 10$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{R_3}$ through R_3 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_3}$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_1} = \frac{10^9}{s + 2 \times 10^5}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_3} = \frac{-10^5}{s + 2 \times 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 0.2685$, so $|H_2| = 10 \times 0.2685 = 2.685$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	10	$10A$	$3.142 \times 10^6 A$
V_o	2.685	$2.685A$	$8.436 \times 10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_{o1}, reaches the limit first.

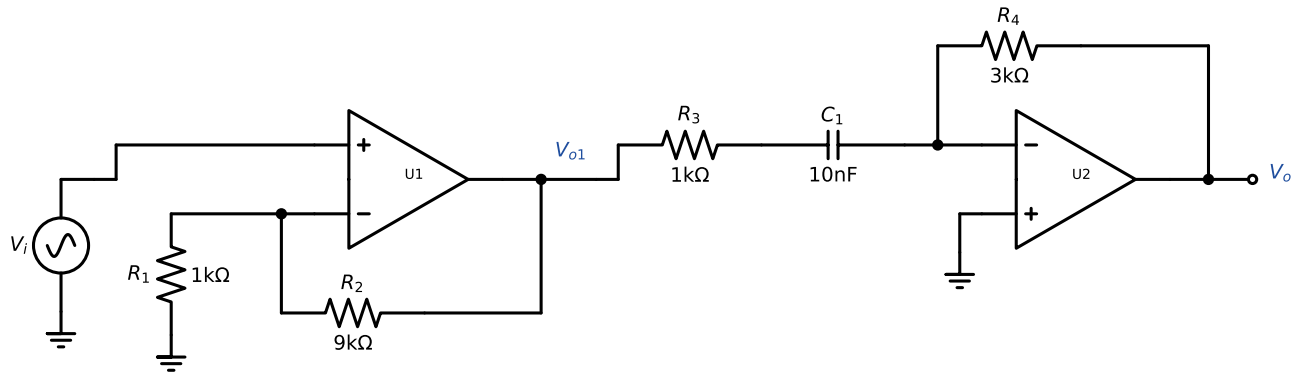
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{10^6}{3.14\times10^5\times10} = 318\,\text{mV}$$

Answer

V_{o1} distorts first. $A_{max} = 318\,\text{mV}$ (input amplitude); at that point V_{o1} has amplitude $3.18\,\text{V}$.

Solution 2E.8 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 3000 = 18800\,\text{rad/s}$ and $SR = 0.5\,\text{V}/\mu\text{s} = 5 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The op-amp forces $V_- = V_+$, and R_1, R_2 divide the output: $V_- = \frac{R_1}{R_1+R_2}V_{o1}$, hence $V_{o1} = \left(1 + \frac{R_2}{R_1}\right)V_+$.

$$\frac{V_{o1}}{V_i} = \left(1 + \frac{R_2}{R_1}\right) = 10$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 10$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{Z_{in2}}$ through Z_{in2} cannot enter the op-amp, so all of it flows through R_4: $V_o = 0 - R_4 \cdot \frac{V_{o1}}{Z_{in2}}$.

$$Z_{in2} = R_3 + \frac{1}{sC_1} = \frac{1000\,s + 10^8}{s}$$

$$\frac{V_o}{V_{o1}} = -\frac{R_4}{Z_{in2}} = \frac{-3\,s}{s + 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 0.5557$, so $|H_2| = 10 \times 0.5557 = 5.557$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	10	$10A$	$1.885 \times 10^5 A$
V_o	5.557	$5.557A$	$1.047 \times 10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_{o1}, reaches the limit first.

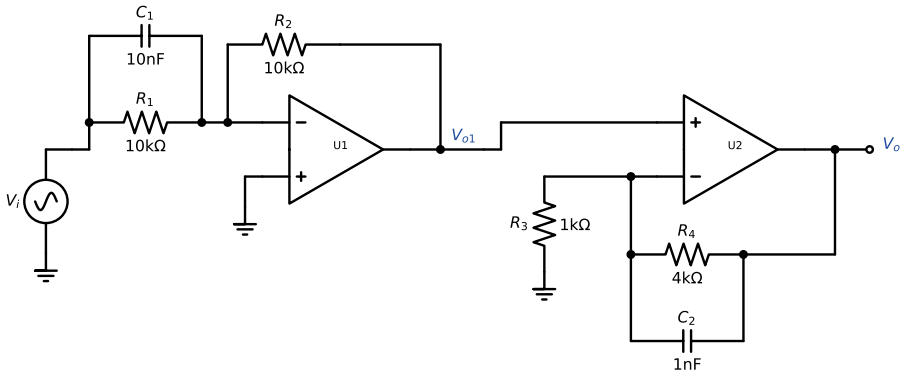
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{5\times10^{5}}{18800\times10} = 2.65\,\text{V}$$

Answer

V_{o1} distorts first. $A_{max} = 2.65\,\text{V}$ (input amplitude); at that point V_{o1} has amplitude $26.5\,\text{V}$.

Solution 2E.9 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 10000 = 62800\,\text{rad/s}$ and $SR = 1\,\text{V}/\mu\text{s} = 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_{o1} = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 \parallel \frac{1}{sC_1} = \frac{10^8}{s + 10000}$$

$$\frac{V_{o1}}{V_i} = -\frac{R_2}{Z_{in1}} = -10^{-4}\,s - 1$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 6.362$.

Block 2 (U2). Non-inverting block: The + input is the block input: $V_+ = V_{o1}$. The op-amp forces $V_- = V_+$, and R_3, Z_{f2} divide the output: $V_- = \frac{R_3}{R_3 + Z_{f2}}V_o$, hence $V_o = \left(1 + \frac{Z_{f2}}{R_3}\right)V_+$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 2.5 \times 10^5}$$

$$\frac{V_o}{V_{o1}} = \left(1 + \frac{Z_{f2}}{R_3}\right) = \frac{s + 1.25 \times 10^6}{s + 2.5 \times 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 4.855$, so $|H_2| = 6.362 \times 4.855 = 30.89$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	6.362	$6.362A$	$3.998 \times 10^5 A$
V_o	30.89	$30.89A$	$1.941 \times 10^6 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_o, reaches the limit first.

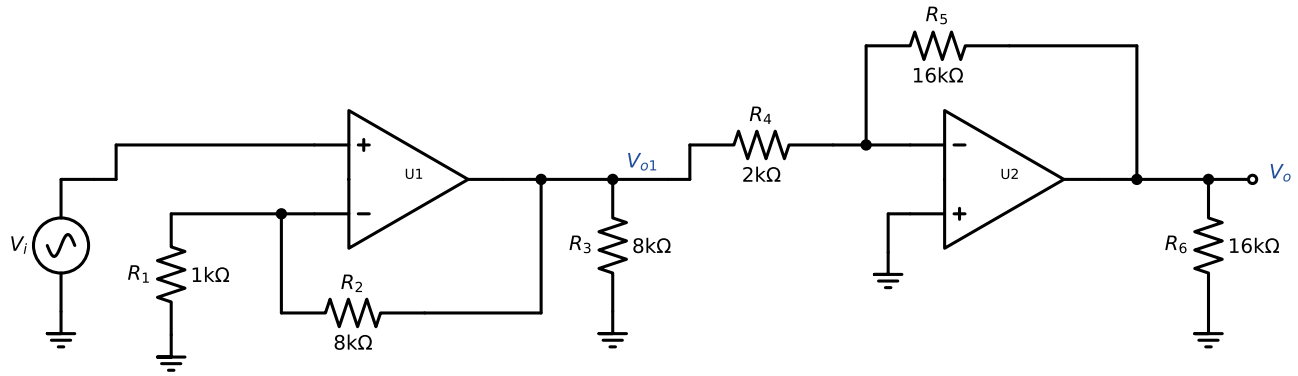
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{10^6}{62800 \times 30.89} = 515\,\text{mV}$$

Answer

V_o distorts first. $A_{max} = 515\,\text{mV}$ (input amplitude); at that point V_o has amplitude $15.9\,\text{V}$.

Solution 2E.10 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 2 \times 10^5 = 1.26 \times 10^6$ rad/s and $SR = 2\,\text{V}/\mu\text{s} = 2 \times 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Each op-amp has $A(s) = \frac{10^5}{1+s/(2\pi\cdot 20)}$, so each block is $\frac{A(\alpha_+-\alpha_-)}{1+A\beta}$ (tool T5; α, β from dividers, as in the finite-gain set).

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1+R_2}V_{o1}$$
$$\frac{V_{o1}}{V_i} = \frac{A(\alpha_+-\alpha_-)}{1+A\beta} = \frac{1.26\times 10^7}{s+1.4\times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 6.689$.

Block 2 (U2). (superposition: V_{o1} alone with $V_o = 0$ gives a divider R_4–R_5, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_5}{R_4+R_5}V_{o1} + \frac{R_4}{R_4+R_5}V_o$$
$$\frac{V_o}{V_{o1}} = \frac{A(\alpha_+-\alpha_-)}{1+A\beta} = \frac{-1.12\times 10^7}{s+1.4\times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 5.946$, so $|H_2| = 6.689 \times 5.946 = 39.78$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a+jb| = \sqrt{a^2+b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	6.689	$6.689A$	$8.406\times 10^6 A$
V_o	39.78	$39.78A$	$4.998\times 10^7 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_o, reaches the limit first.

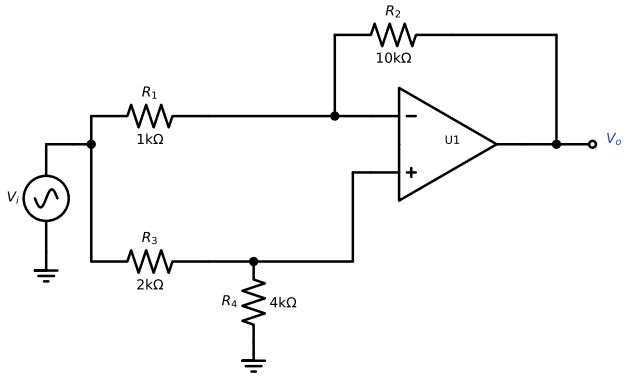
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{2\times10^{6}}{1.26\times10^{6}\times39.78} = 40\,\text{mV}$$

Answer

V_o distorts first. $A_{max} = 40\,\text{mV}$ (input amplitude); at that point V_o has amplitude $1.59\,\text{V}$.

Solution 2E.11 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 50000 = 3.14 \times 10^5$ rad/s and $SR = 3\,\text{V}/\mu\text{s} = 3 \times 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Each op-amp has $A(s) = \frac{10^5}{1+s/(2\pi\cdot 10)}$, so each block is $\frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$ (tool T5; α, β from dividers, as in the finite-gain set).

Block 1 (U1). (each input is a divider; the − input sees both sources, use superposition)

$$V_+ = \frac{R_4}{R_3 + R_4}V_i$$

$$V_- = \frac{R_2}{R_1 + R_2}V_i + \frac{R_1}{R_1 + R_2}V_o$$

$$\frac{V_o}{V_i} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{-1.52 \times 10^6}{s + 5.71 \times 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 2.336$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	2.336	$2.336A$	$7.34 \times 10^5 A$

There is only one op-amp output.

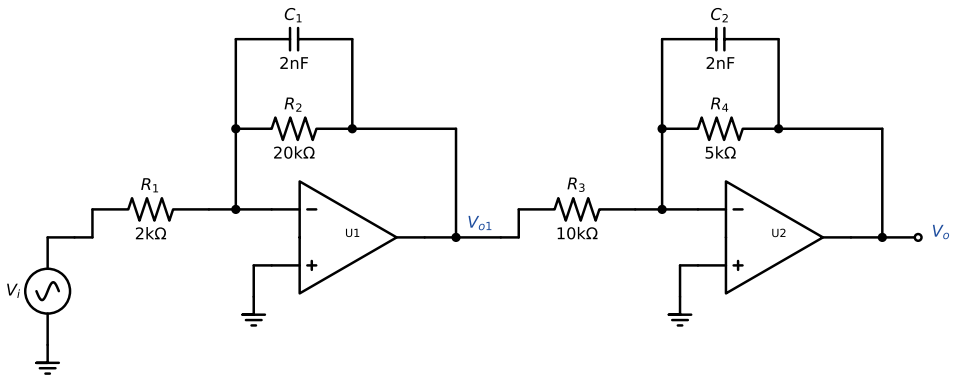
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{3 \times 10^6}{3.14 \times 10^5 \times 2.336} = 4.09\,\text{V}$$

Answer

$A_{max} = 4.09\,\text{V}$ (input amplitude); at that point V_o has amplitude $9.55\,\text{V}$.

Solution 2E.12 **Medium**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 8000 = 50300\,\text{rad/s}$ and $SR = 0.5\,\text{V}/\mu\text{s} = 5 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{R_1}$ through R_1 cannot enter the op-amp, so all of it flows through Z_{f1}: $V_{o1} = 0 - Z_{f1} \cdot \frac{V_i}{R_1}$.

$$Z_{f1} = R_2 \parallel \frac{1}{sC_1} = \frac{5 \times 10^8}{s + 25000}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-2.5 \times 10^5}{s + 25000}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 4.453$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{R_3}$ through R_3 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_3}$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_2} = \frac{5 \times 10^8}{s + 10^5}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_3} = \frac{-50000}{s + 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 0.4467$, so $|H_2| = 4.453 \times 0.4467 = 1.989$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	4.453	$4.453A$	$2.238 \times 10^5 A$
V_o	1.989	$1.989A$	$100000A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_{o1}, reaches the limit first.

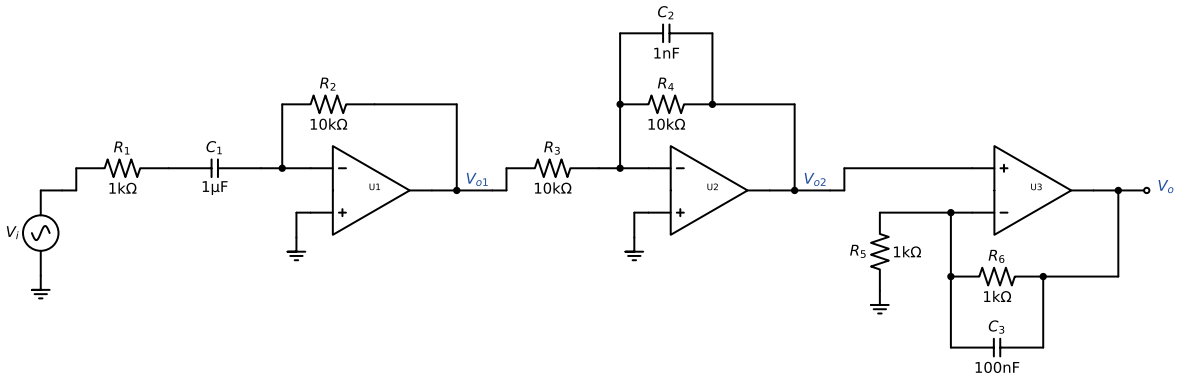
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{5\times10^{5}}{50300\times4.453} = 2.23\,\text{V}$$

Answer

V_{o1} distorts first. $A_{max} = 2.23\,\text{V}$ (input amplitude); at that point V_{o1} has amplitude $9.95\,\text{V}$.

Solution 2E.13 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_{o2}, V_o). Here $\omega_{in} = 2\pi \cdot 20000 = 1.26 \times 10^5$ rad/s and $SR = 1\,\text{V}/\mu\text{s} = 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through R_2: $V_{o1} = 0 - R_2 \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^6}{s}$$

$$\frac{V_{o1}}{V_i} = -\frac{R_2}{Z_{in1}} = \frac{-10\,s}{s + 1000}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 10$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{R_3}$ through R_3 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_{o2} = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_3}$.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 10^5}$$

$$\frac{V_{o2}}{V_{o1}} = -\frac{Z_{f2}}{R_3} = \frac{-10^5}{s + 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o2}}{V_{o1}}\right| = 0.6227$, so $|H_2| = 10 \times 0.6227 = 6.227$.

Block 3 (U3). Non-inverting block: The + input is the block input: $V_+ = V_{o2}$. The op-amp forces $V_- = V_+$, and R_5, Z_{f3} divide the output: $V_- = \frac{R_5}{R_5 + Z_{f3}} V_o$, hence $V_o = \left(1 + \frac{Z_{f3}}{R_5}\right) V_+$.

$$Z_{f3} = R_6 \parallel \frac{1}{sC_3} = \frac{10^7}{s + 10000}$$

$$\frac{V_o}{V_{o2}} = \left(1 + \frac{Z_{f3}}{R_5}\right) = \frac{s + 20000}{s + 10000}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o2}}\right| = 1.009$, so $|H_3| = 6.227 \times 1.009 = 6.285$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	10	$10A$	$1.257 \times 10^6 A$
V_{o2}	6.227	$6.227A$	$7.825 \times 10^5 A$
V_o	6.285	$6.285A$	$7.898 \times 10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_{o1}, reaches the limit first.

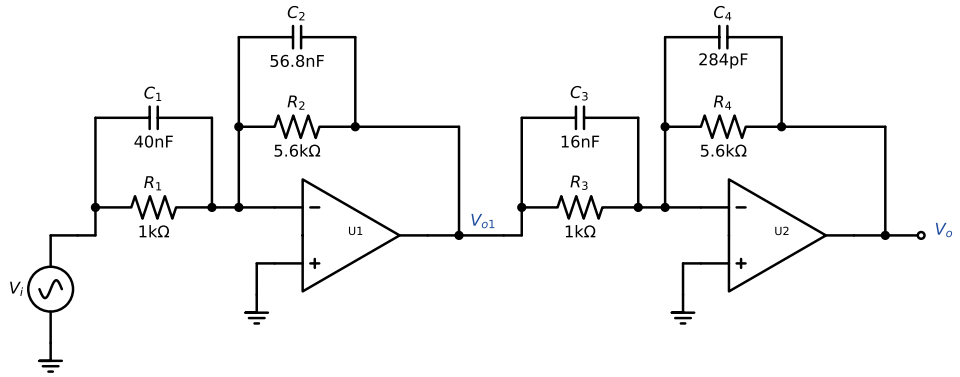
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{10^6}{1.26 \times 10^5 \times 10} = 796\,\text{mV}$$

Answer

V_{o1} distorts first. $A_{max} = 796\,\text{mV}$ (input amplitude); at that point V_{o1} has amplitude $7.96\,\text{V}$.

Solution 2E.14 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 2000 = 12600\,\text{rad/s}$ and $SR = 0.5\,\text{V}/\mu\text{s} = 5 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_i - 0}{Z_{in1}}$ through Z_{in1} cannot enter the op-amp, so all of it flows through Z_{f1}: $V_{o1} = 0 - Z_{f1} \cdot \frac{V_i}{Z_{in1}}$.

$$Z_{in1} = R_1 \parallel \frac{1}{sC_1} = \frac{2.5 \times 10^7}{s + 25000}$$
$$Z_{f1} = R_2 \parallel \frac{1}{sC_2} = \frac{1.76 \times 10^7}{s + 3140}$$
$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{Z_{in1}} = \frac{-0.704\,s - 17600}{s + 3140}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 1.521$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1} - 0}{Z_{in2}}$ through Z_{in2} cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{Z_{in2}}$.

$$Z_{in2} = R_3 \parallel \frac{1}{sC_3} = \frac{6.25 \times 10^7}{s + 62500}$$
$$Z_{f2} = R_4 \parallel \frac{1}{sC_4} = \frac{3.52 \times 10^9}{s + 6.29 \times 10^5}$$
$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{Z_{in2}} = \frac{-56.3\,s - 3.52 \times 10^6}{s + 6.29 \times 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 5.711$, so $|H_2| = 1.521 \times 5.711 = 8.687$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	1.521	$1.521A$	$19120A$
V_o	8.687	$8.687A$	$1.092 \times 10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_o, reaches the limit first.

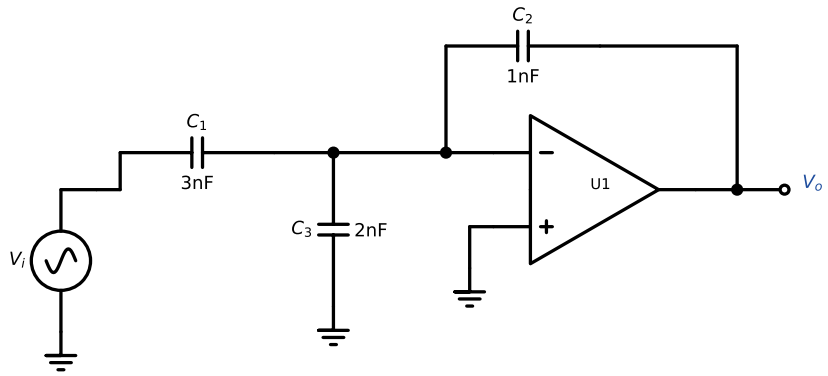
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{5 \times 10^5}{12600 \times 8.687} = 4.58\,\text{V}$$

Answer

V_o distorts first. $A_{max} = 4.58\,\text{V}$ (input amplitude); at that point V_o has amplitude 39.8 V.

Solution 2E.15 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to the output. Here $\omega_{in} = 2\pi \cdot 5 \times 10^5 = 3.14 \times 10^6\,\text{rad/s}$ and $SR = 5\,\text{V}/\mu\text{s} = 5 \times 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Each op-amp has $A(s) = \frac{10^5}{1+s/(2\pi\cdot 30)}$, so each block is $\frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$ (tool T5; α, β from dividers, as in the finite-gain set).

Block 1 (U1). (superposition: with $V_o = 0$, $\frac{1}{sC_2}$ and $\frac{1}{sC_3}$ are in parallel from the $-$ input to ground; with $V_i = 0$, $\frac{1}{sC_1}$ and $\frac{1}{sC_3}$ are. $\frac{1}{sC_3}$ matters now, because V_- is no longer forced to 0.)

$$V_+ = 0 \quad \text{(grounded)}$$

$$V_- = \frac{\frac{1}{sC_2} \parallel \frac{1}{sC_3}}{\frac{1}{sC_1} + \frac{1}{sC_2} \parallel \frac{1}{sC_3}} V_i + \frac{\frac{1}{sC_1} \parallel \frac{1}{sC_3}}{\frac{1}{sC_2} + \frac{1}{sC_1} \parallel \frac{1}{sC_3}} V_o$$

$$\frac{V_o}{V_i} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{-9.42 \times 10^6}{s + 3.14 \times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_i}\right| = 2.121$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_o	2.121	$2.121A$	$6.664 \times 10^6 A$

There is only one op-amp output.

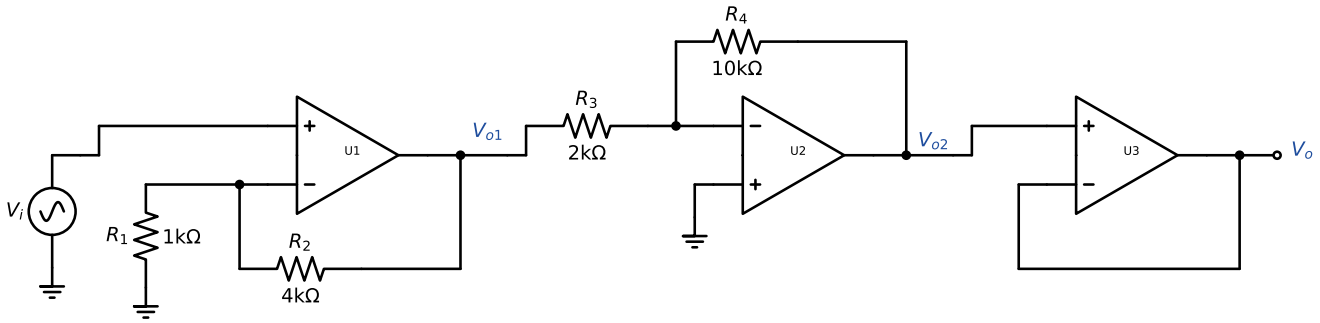
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{5 \times 10^6}{3.14 \times 10^6 \times 2.121} = 750\,\text{mV}$$

Answer

$A_{max} = 750\,\text{mV}$ (input amplitude); at that point V_o has amplitude $1.59\,\text{V}$.

Solution 2E.16 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_{o2}, V_o). Here $\omega_{in} = 2\pi \cdot 10^5 = 6.28 \times 10^5$ rad/s and $SR = 2\,\text{V}/\mu\text{s} = 2 \times 10^6\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Each op-amp has $A(s) = \frac{10^5}{1+s/(2\pi\cdot 10)}$, so each block is $\frac{A(\alpha_+ - \alpha_-)}{1+A\beta}$ (tool T5; α, β from dividers, as in the finite-gain set).

Block 1 (U1). (no current enters either input, so both are plain dividers)

$$V_+ = V_i$$
$$V_- = \frac{R_1}{R_1 + R_2} V_{o1}$$

$$\frac{V_{o1}}{V_i} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{6.28 \times 10^6}{s + 1.26 \times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 4.472$.

Block 2 (U2). (superposition: V_{o1} alone with $V_{o2} = 0$ gives a divider R_3–R_4, and vice versa)

$$V_+ = 0 \quad \text{(grounded)}$$
$$V_- = \frac{R_4}{R_3 + R_4} V_{o1} + \frac{R_3}{R_3 + R_4} V_{o2}$$

$$\frac{V_{o2}}{V_{o1}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{-5.24 \times 10^6}{s + 1.05 \times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o2}}{V_{o1}}\right| = 4.287$, so $|H_2| = 4.472 \times 4.287 = 19.17$.

Block 3 (U3). (no current enters either input, so both are plain dividers)

$$V_+ = V_{o2}$$
$$V_- = V_o$$

$$\frac{V_o}{V_{o2}} = \frac{A(\alpha_+ - \alpha_-)}{1 + A\beta} = \frac{6.28 \times 10^6}{s + 6.28 \times 10^6}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o2}}\right| = 0.995$, so $|H_3| = 19.17 \times 0.995 = 19.08$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	4.472	$4.472A$	$2.81 \times 10^6 A$
V_{o2}	19.17	$19.17A$	$1.205 \times 10^7 A$
V_o	19.08	$19.08A$	$1.199 \times 10^7 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_{o2}, reaches the limit first.

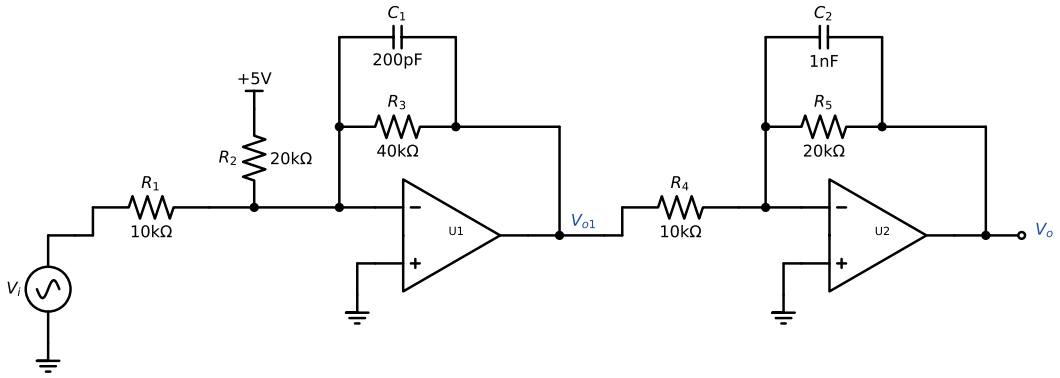
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{2 \times 10^6}{6.28 \times 10^5 \times 19.17} = 166\,\text{mV}$$

Answer

V_{o2} distorts first. $A_{max} = 166\,\text{mV}$ (input amplitude); at that point V_{o2} has amplitude $3.18\,\text{V}$.

Solution 2E.17 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_o). Here $\omega_{in} = 2\pi \cdot 10000 = 62800\,\text{rad/s}$ and $SR = 0.2\,\text{V}/\mu\text{s} = 2 \times 10^5\,\text{V/s}$. (A DC offset does not change the slope of a signal, so only the AC part matters.)

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Summing block: $V_- = V_+ = 0$ (virtual ground), so the input currents $\frac{V_i}{R_1}$ and $\frac{V_{DC}}{R_2}$ simply add and both flow through Z_{f1}: $V_{o1} = -Z_{f1}\left(\frac{V_i}{R_1} + \frac{V_{DC}}{R_2}\right)$. For the AC part the supply is zero, so R_2 (from virtual ground to AC ground) carries no AC current and drops out.

$$Z_{f1} = R_3 \parallel \frac{1}{sC_1} = \frac{5 \times 10^9}{s + 1.25 \times 10^5}$$

$$\frac{V_{o1}}{V_i} = -\frac{Z_{f1}}{R_1} = \frac{-5 \times 10^5}{s + 1.25 \times 10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 3.574$.

Block 2 (U2). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o1}-0}{R_4}$ through R_4 cannot enter the op-amp, so all of it flows through Z_{f2}: $V_o = 0 - Z_{f2} \cdot \frac{V_{o1}}{R_4}$.

$$Z_{f2} = R_5 \parallel \frac{1}{sC_2} = \frac{10^9}{s + 50000}$$

$$\frac{V_o}{V_{o1}} = -\frac{Z_{f2}}{R_4} = \frac{-10^5}{s + 50000}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o1}}\right| = 1.245$, so $|H_2| = 3.574 \times 1.245 = 4.451$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	3.574	$3.574A$	$2.246 \times 10^5 A$
V_o	4.451	$4.451A$	$2.797 \times 10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_o, reaches the limit first.

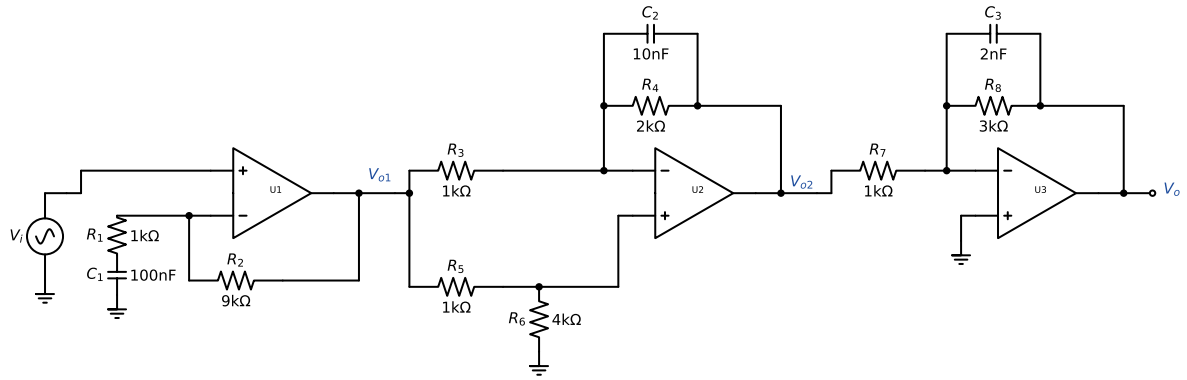
Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{2\times10^5}{62800 \times 4.451} = 715\,\text{mV}$$

Answer

V_o distorts first. $A_{max} = 715\,\text{mV}$ (input amplitude); at that point V_o has amplitude $3.18\,\text{V}$.

Solution 2E.18 **Hard**



Step 1 — What limits the input?

Every op-amp output must obey tool T6. For $V_i = A\sin(\omega_{in}t)$ each op-amp output is a sinusoid $|H_k(j\omega_{in})|A\sin(\omega_{in}t + \dots)$, so its fastest slope is $\omega_{in}|H_k(j\omega_{in})|A$. We need $|H_k(j\omega_{in})|$ from V_i to each op-amp output (V_{o1}, V_{o2}, V_o). Here $\omega_{in} = 2\pi \cdot 6000 = 37700\,\text{rad/s}$ and $SR = 0.8\,\text{V}/\mu\text{s} = 8 \times 10^5\,\text{V/s}$.

Step 2 — Gain to every op-amp output at ω_{in}

Ideal op-amps: each block follows from its block formula (tool T2); a cascade multiplies, so $|H_k| = \prod_{j\le k}|\text{block } j|$.

Block 1 (U1). Non-inverting block: The + input is the block input: $V_+ = V_i$. The op-amp forces $V_- = V_+$, and Z_{g1}, R_2 divide the output: $V_- = \frac{Z_{g1}}{Z_{g1}+R_2}V_{o1}$, hence $V_{o1} = \left(1 + \frac{R_2}{Z_{g1}}\right)V_+$.

$$Z_{g1} = R_1 + \frac{1}{sC_1} = \frac{1000\,s + 10^7}{s}$$

$$\frac{V_{o1}}{V_i} = \left(1 + \frac{R_2}{Z_{g1}}\right) = \frac{10\,s + 10000}{s + 10000}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o1}}{V_i}\right| = 9.669$.

Block 2 (U2). Difference block (both inputs driven by V_{o1}): use superposition. Through the + path the signal is divided, $V_+ = \frac{R_6}{R_5+R_6}V_{o1}$, and amplified by the non-inverting gain $1 + \frac{Z_{f2}}{R_3}$; through the − path it is amplified by the inverting gain $-\frac{Z_{f2}}{R_3}$. Add the two contributions.

$$Z_{f2} = R_4 \parallel \frac{1}{sC_2} = \frac{10^8}{s + 50000}$$

$$\frac{V_{o2}}{V_{o1}} = \frac{R_6}{R_5 + R_6}\left(1 + \frac{Z_{f2}}{R_3}\right) - \frac{Z_{f2}}{R_3} = \frac{0.8\,s + 20000}{s + 50000}$$

At $s = j\omega_{in}$: $\left|\frac{V_{o2}}{V_{o1}}\right| = 0.5779$, so $|H_2| = 9.669 \times 0.5779 = 5.588$.

Block 3 (U3). Inverting block: the + input is grounded, so $V_- = V_+ = 0$ (virtual ground). The current $\frac{V_{o2}-0}{R_7}$ through R_7 cannot enter the op-amp, so all of it flows through Z_{f3}: $V_o = 0 - Z_{f3} \cdot \frac{V_{o2}}{R_7}$.

$$Z_{f3} = R_8 \parallel \frac{1}{sC_3} = \frac{5 \times 10^8}{s + 1.67 \times 10^5}$$

$$\frac{V_o}{V_{o2}} = -\frac{Z_{f3}}{R_7} = \frac{-5\times10^5}{s+1.67\times10^5}$$

At $s = j\omega_{in}$: $\left|\frac{V_o}{V_{o2}}\right| = 2.926$, so $|H_3| = 5.588 \times 2.926 = 16.35$.

(To evaluate a magnitude, put $s = j\omega_{in}$ and use $|a + jb| = \sqrt{a^2 + b^2}$ for the numerator and the denominator separately.)

Step 3 — Which output slews first?

output	$\lvert H_k(j\omega_{in})\rvert$	amplitude	max slope $\omega_{in}\lvert H_k\rvert A$ (V/s)
V_{o1}	9.669	$9.669A$	$3.645\times10^5 A$
V_{o2}	5.588	$5.588A$	$2.107\times10^5 A$
V_o	16.35	$16.35A$	$6.164\times10^5 A$

All op-amps have the same slew rate, so the output with the *largest* gain, V_o, reaches the limit first.

Step 4 — Maximum input amplitude

$$\omega_{in}\,|H|\,A \le SR \;\Rightarrow\; A_{max} = \frac{SR}{\omega_{in}|H|} = \frac{8\times10^5}{37700\times16.35} = 1.3\,\text{V}$$

Answer

V_o distorts first. $A_{max} = 1.3\,\text{V}$ (input amplitude); at that point V_o has amplitude 21.2 V.